

CNML Poetry Inline DSL + Geometry Spec (v0.1)

Core Concepts:

Anchors are zero-dimensional points. Spans are one-dimensional extents.

All geometry is computed post-layout using exact rational arithmetic.

Syntax:

Anchors: `{{@name(:role)*}}`

Spans: `{{#name(:role)* :: content}}`

Placement: `{{expression :: content}}`

Roles:

Core roles: beat, stress, pause, caesura

Multiple roles allowed per element. Roles are unordered and deduplicated.

Naming:

Names are fully qualified and scoped across the entire .

Nested elements inherit parent names using dot notation (e.g., `s1.a`, `s1.end`).

Geometry:

Span `S` defines: left, right, center, width

$\text{width} = \text{right} - \text{left}$; $\text{center} = (\text{left} + \text{right})/2$

Expressions:

Form: `(+|-) *`

`=` anchor

`=` rational OR rational * span.width (or span shorthand)

Units:

Units are non-rendering spans with role 'measure'.

Example: `{{#em:measure :: M}}`

In length expressions, 'em' resolves to `em.width`.

hspace:

provides local cursor-based spacing.

It does not participate in global geometry or dependency graphs.